

Mains Supply Interruption Alarm



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This inexpensive circuit can be used as a mains supply interruption alarm. When mains supply gets interrupted, it gives an audible warning for a few seconds.

Circuit and working

Circuit diagram of the mains supply interruption alarm is shown in Fig. 1. It is built around step-down transformer X1, two 1N4007 diodes (D1 and D2), two BC547 transistors (T1 and T2), a piezo buzzer (PZ1) and a few other components.

PARTS LIST

Semiconductors:

- T1, T2 - BC547 npn transistor
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, 45% carbon):

- R1 - 47-kilo-ohm
- R2 - 22-kilo-ohm

Capacitors:

- C1 - 4.7µF, 35V electrolytic
- C2, C3 - 2200µF, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- X1 - 230V AC primary to 12V, 200mA secondary transformer
- PZ1 - Piezo buzzer

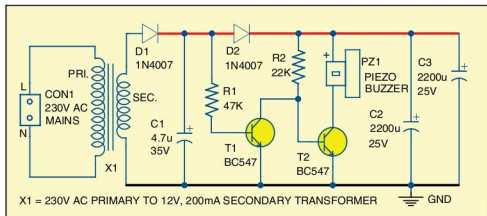


Fig. 1: Circuit diagram of the mains supply interruption indicator

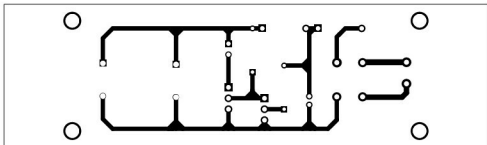


Fig. 2: Actual-size PCB of the mains supply interruption indicator

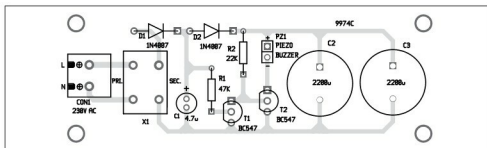


Fig. 3: Component layout of the PCB

In presence of 230V AC mains supply, transistor T1 conducts, transistor T2 does not conduct and so the piezo buzzer does not make any sound. During this period, capacitors C2 and C3 get charged.

When 230V AC mains supply interrupts, transistor T1 does not conduct, transistor T2 conducts due to the charge obtained from the two 2200µF capacitors and the piezo buzzer sounds for a few seconds.

The circuit remains silent when 230V AC mains power supply is available and sounds only when the mains gets interrupted.

Construction and testing

An actual-size, single-side PCB for the mains supply interruption indicator is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a small box so that 230V AC mains can be connected easily.

EFY note. Use only a piezo buzzer and not a higher-current-drawing coil-type buzzer. ●



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